Learning to Stop: Process-Signal Early Exit for Diffusion Models

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Abstract

Text-to-image diffusion models run a fixed step budget per sample even though individual prompts converge at very different rates. We investigate whether the U-Net's own internal state, read out at a handful of checkpoints, carries enough information to decide *per image* when further denoising is no longer worthwhile. We introduce SignalStop, a zero-overhead early-exit policy: at each checkpoint we extract ten interpretable scalar signals from quantities the U-Net already computes and feed them to a single logistic regression that predicts stop/continue. Labels are derived offline from PickScore convergence; at inference the policy touches neither a reward model nor any auxiliary network. Evaluated under strict prompt-grouped cross-validation on 2419 prompts pooled from PartiPrompts, MS-COCO, and DrawBench, SignalStop achieves ROC-AUC 0.933 $[0.929, 0.937]$ and F_1(STOP) 0.796 $[0.788, 0.805]$ at the recommended operating point τ=0.80, saving 32.6% of denoising compute on average; the 0.80 F_1 target sits inside the 95% bootstrap CI. Performance is corpus-robust (AUC 0.92–0.94 across the three sources), and a slice of the pooled model restricted to the original 1000-prompt PartiPrompts cohort recovers the standalone-trained baseline to within AUC 0.0002, confirming that scaling does not dilute the signal. A leave-one-group-out analysis attributes essentially all of the lift over a step-position-only baseline to a single signal group, the classifier-free-guidance gap: removing it drops AUC by 0.025; removing any other group changes AUC by at most 0.001. Applied zero-shot to DDIM and DPM-Solver++ on a paired 500-prompt subset, the frozen classifier transfers at AUC deltas of -0.003 and -0.048 respectively, with a one-scalar-per-scheduler threshold recalibration recovering sensible policy behaviour on both targets. SignalStop thus identifies a compact, interpretable internal probe for convergence, usable as an adaptive early-exit policy with no modification to the diffusion model.

1 Introduction

Text-to-image diffusion models [3, 9, 10] iteratively denoise a Gaussian latent over a fixed number of steps—typically 20–50. The schedule is set once and applied uniformly: every prompt, regardless of difficulty, pays the same compute bill. This is evidently wasteful, and the waste is not uniform. Compositional prompts with long descriptions, fine text, or multi-object spatial relations continue gaining quality through the final steps; short, prototypical prompts stabilise long before the run ends. A sampler that could identify, on the fly, which of these regimes a given prompt is in would deliver compute savings proportional to the fraction of the corpus that converges early—without global schedule changes or retraining.

A substantial body of work targets this through *global* step reduction: progressive distillation [8, 12], consistency models [7, 14], higher-order solvers [4, 6]. All apply a single schedule to every prompt and cannot differentiate easy inputs from hard ones. A smaller line of work

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adapts *per input*: AdaDiff [16] trains a separate RL policy to select skip patterns; DeeDiff [17] inserts trained uncertainty heads at each U-Net block and exits early based on their outputs. Both require auxiliary networks and additional training.

We take a different starting point. The U-Net at step t already produces the noise prediction, the CFG delta, the predicted clean latent $\hat{x}_0$, and cross-attention maps. If the model itself is close to converged, these quantities should reflect that. We therefore ask:

> *Can a simple classifier, reading only scalar summaries of the U-Net's existing per-step outputs, decide when to stop on a per-image basis?*

Contributions.

1. We define ten interpretable scalar signals (Sec. 3) computable from a single SDXL forward pass, with no auxiliary network and negligible per-step overhead.

2. We introduce SignalStop (Sec. 4), a per-image early-exit policy that feeds these signals to a single logistic regression trained on PickScore-derived convergence labels. The policy touches no reward model at inference.

3. Evaluated on 2419 prompts drawn from PartiPrompts, MS-COCO and DrawBench (Sec. 6), SignalStop reaches ROC-AUC 0.933 [0.929, 0.937] and saves 32.6% of denoising compute at τ=0.80, with AUC 0.92–0.94 across all three corpora and a +0.0002 AUC replication on the 1000-prompt subset shared with our earlier single-corpus study.

4. A leave-one-group-out analysis (Sec. 7) isolates the CFG-gap pair as the dominant discriminative signal (ΔAUC = -0.025 when removed; all other groups $|\Delta| < 0.001$). The effective predictor is a two-feature CFG-gap probe on top of step position—a compact mechanistic finding rather than a multi-signal black box.

5. We test cross-scheduler transfer (Sec. 6.6). The Euler-trained classifier, applied zero-shot to DPM-Solver++ and DDIM trajectories on a paired 500-prompt subset, retains per-corpus AUC in the 0.86–0.93 band and transfers cleanly to DDIM (ΔAUC -0.003); DPM-Solver++ requires a one-scalar threshold recalibration to control its operational policy, consistent with that scheduler's compressed 20-step convergence profile.

2 Related Work

Global step reduction. Progressive distillation [12] and consistency models [7, 14] compress the sampling trajectory into a small fixed number of steps. DPM-Solver [6] and EDM [4] achieve similar reductions through better numerical integration. All these methods apply the same step budget to every prompt.

Adaptive step selection. AdaDiff [16] selects skip patterns via reinforcement learning on a separate policy network. DeeDiff [17] adds trained uncertainty heads to each U-Net block and exits early based on their outputs, requiring architectural modification and joint training. Align Your Steps [13] optimises the *schedule* (step positions) once globally, not the per-prompt stopping decision. SignalStop differs from all three: no auxiliary network, no architectural modification, and no retraining of the diffusion model. It reads scalar summaries of what the U-Net already computes in a standard forward pass and makes a per-image stop decision with a single linear model.

Probing U-Net internals. Cross-attention maps [1, 2] and intermediate latents have previously been used for editing and grounding [15]. We use the same quantities for a different purpose—monitoring convergence.

Quality measurement. PickScore [5] provides a differentiable human-preference proxy. We use it only offline, to produce training labels for SignalStop; the inference-time policy never calls it.

3 Process Signals

At every checkpoint $t \in \{5, 10, 15, 20, 25, 30\}$ of a 30-step SDXL run we extract ten scalar signals that summarise the current state of the denoising process. All are computed from tensors produced by the standard U-Net forward pass—no additional forward passes, no auxiliary networks. Let $\varepsilon_t^{\text{cond}}$ and $\varepsilon_t^{\text{uncond}}$ denote the conditional and unconditional noise predictions at step t, and let $\hat{x}_{0,t}$ be the predicted clean latent returned by the scheduler.

3.1 Signal definitions

1–2. CFG gap. Classifier-free guidance steers generation via $\varepsilon_t = \varepsilon_t^{\text{uncond}} + w\,(\varepsilon_t^{\text{cond}} - \varepsilon_t^{\text{uncond}})$. We measure the instantaneous and cumulative magnitude of the guidance correction:

$$\text{cfg\_gap\_norm}_t = \|\varepsilon_t^{\text{cond}} - \varepsilon_t^{\text{uncond}}\|_2, \tag{1}$$

$$\text{cfg\_gap\_cumulative}_t = \sum_{s \le t} \text{cfg\_gap\_norm}_s. \tag{2}$$

The per-step norm tracks how much semantic work guidance is still doing; the cumulative version encodes how much it has already done.

3–4. Predicted clean-image stability. The scheduler output $\hat{x}_{0,t}$ is the model's running estimate of the final image. Its rate of change is a direct convergence indicator:

$$\text{x0\_mse}_t = \|\hat{x}_{0,t} - \hat{x}_{0,t-\Delta}\|_2^2, \tag{3}$$

$$\text{x0\_cosine}_t = \frac{\langle \hat{x}_{0,t}, \hat{x}_{0,t-\Delta} \rangle}{\|\hat{x}_{0,t}\|\,\|\hat{x}_{0,t-\Delta}\|}. \tag{4}$$

5–6. Cross-attention concentration. We extract the averaged cross-attention map $A_t \in \mathbb{R}^{HW \times L}$ from the U-Net mid-block and compute

$$\text{cross\_attn\_entropy}_t = -\tfrac{1}{HW} \sum_{i,j} A_{t,ij} \log A_{t,ij}, \tag{5}$$

$$\text{cross\_attn\_max}_t = \tfrac{1}{HW} \sum_{i} \max_j A_{t,ij}. \tag{6}$$

As denoising progresses, attention typically sharpens onto prompt-relevant regions.

7–8. Noise-prediction norm.

$$\text{eps\_norm}_t = \|\varepsilon_t\|_2, \qquad \text{eps\_norm\_delta}_t = \|\varepsilon_t\|_2 - \|\varepsilon_{t-\Delta}\|_2. \tag{7}$$

9–10. High-frequency power. With $F_t = \mathcal{F}\{\hat{x}_{0,t}\}$ the 2-D Fourier transform of the predicted clean latent and $\mathcal{H}$ the frequencies above the median,

$$\text{hf\_power\_ratio}_t = \frac{\sum_{k \in \mathcal{H}} |F_{t,k}|^2}{\sum_k |F_{t,k}|^2}, \quad \text{hf\_power\_delta}_t = \text{hf\_power\_ratio}_t - \text{hf\_power\_ratio}_{t-\Delta}. \tag{8}$$

3.2 Computational overhead

Every signal is a reduction over a tensor the U-Net already produces. The added cost per checkpoint is an FFT on the predicted clean latent plus a few norms and reductions—under 1% of the per-step U-Net runtime at checkpoints, and zero at non-checkpoint steps. Crucially, no additional forward passes, no PickScore calls, and no VAE decoding are required at inference time.

4 Method: SignalStop

4.1 Problem formulation

Given a prompt and a T=30-step diffusion run, let $\mathcal{C} = \{5, 10, 15, 20, 25, 30\}$ be the checkpoint set. At each $t \in \mathcal{C} \setminus \{T\}$ we must decide $a_t \in \{\textsc{stop}, \textsc{continue}\}$. Once STOP is issued, the current $\hat{x}_{0,t}$ is decoded as the final image and the remaining steps are skipped.

4.2 Label generation

Labels come from PickScore convergence. With s_t the PickScore of $\hat{x}_{0,t}$ after VAE decoding, define the per-trajectory *progress*

$$\mathrm{prog}_t = \frac{s_t - s_5}{s_T - s_5} \tag{9}$$

(clipped to $[0, 1]$) and the binary label $y_t = \mathbf{1}[\mathrm{prog}_t < \tau]$, where $y_t = 0$ encodes STOP. Figure 1 shows the progress distribution per step on the full corpus: the median rises steeply through step 20 and then saturates, so any $\tau \in [0.80, 0.90]$ separates "has converged" from "still gaining" with a clear margin. We report a full sweep $\tau \in \{0.80, 0.85, 0.90, 0.95\}$ in Sec. 7; τ=0.80 is our recommended operating point and τ=0.90 is retained as a strict reference that matches our prior, single-corpus baseline.

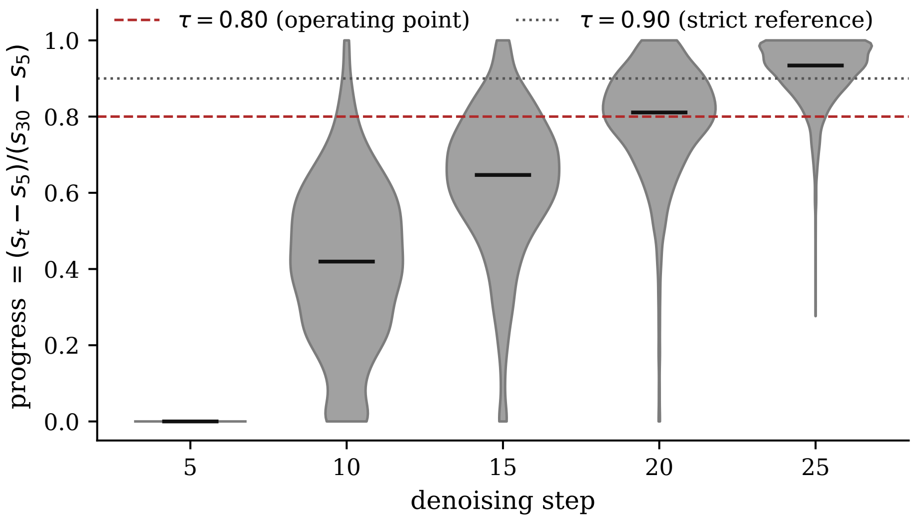

Figure 1: Progress $(s_t - s_5)/(s_T - s_5)$ distribution per checkpoint on the full 2419-prompt corpus (violin; median marked). The two horizontal lines mark the label thresholds τ=0.80 (operating point, red) and τ=0.90 (strict reference, grey).

Degenerate trajectories. A small fraction of trajectories have near-flat PickScore ($s_T - s_5 < 0.5$)—the model either saturated early or failed to improve. We exclude these rather than admit noisy labels (80 of 2499 prompts, 3.2%). Sensitivity to this filter is bounded: removing it changes AUC by less than 0.006 on the smaller v9 cohort.

Why PickScore for labels, not at inference. PickScore supplies a principled, reproducible definition of "done" that is prompt- and model-aware. It is used *only offline*, to generate training labels. At inference SignalStop sees only the ten scalar signals; it never calls PickScore and never VAE-decodes until the stop step.

4.3 Classifier and inference

We train a single logistic regression on standardised signal vectors, shared across all checkpoints (t itself is an optional additional feature). Class weights are balanced to correct for the step-dependent STOP prior. At inference the classifier runs at each checkpoint; the first t for which it predicts STOP halts the loop and the VAE decodes $\hat{x}_{0,t}$ as the output image. If no checkpoint triggers, the model runs the full T steps.

5 Experimental Setup

Model. Stable Diffusion XL 1.0 [9], fp16 on a single NVIDIA A100 40 GB. Euler discrete scheduler, 30 inference steps, guidance scale w=7.5, resolution 1024×1024, single seed (42).

Corpus. 2499 prompts drawn with seed 42 from three sources: (i) 1632 PartiPrompts [18] (our prior 1000-prompt stratified sample plus 632 of the remaining unused prompts); (ii) 667 MS-COCO captions (Karpathy split, first caption per image); (iii) 200 DrawBench prompts comprising the full published benchmark (the compositional stress-test set released with Imagen [11]). After filtering degenerate trajectories (Sec. 4.2), 2419 prompts remain. Corpora are pooled for training and evaluation; per-corpus slices of the pooled classifier's out-of-fold predictions are reported in Sec. 6.4.

Checkpoints. $\{5, 10, 15, 20, 25, 30\}$; the first five are candidate stop points and step 30 is the reference end-of-run PickScore.

Cross-validation. 5-fold `StratifiedGroupKFold` grouped by `prompt_id` with `random_state=42`. Features are standardised per training fold. All reported metrics are out-of-fold (OOF) aggregates computed on predictions the model has not trained on for any fold, so no form of prompt-level leakage is possible.

Bootstrap. For the headline metrics we report 95% confidence intervals from 500 *cluster* bootstrap resamples: each resample redraws 2419 prompt-ids with replacement (preserving the grouping structure of the data) and re-runs the full 5-fold training and evaluation. CIs are the 2.5–97.5 percentiles of the resulting distribution over resamples.

Baselines.

- **No-stop**: run all 30 steps (full-compute quality upper bound).
- **Oracle**: stop at the first checkpoint with ground-truth label STOP (oracle upper bound on savings given τ).
- **Step-only (LR)**: logistic regression with only t as a feature. The critical baseline: if the signals are not adding information beyond step position, step-only should match SignalStop.
- **Signals + step (LR)**: the ten signals plus t, to test complementarity with step position.

6 Results

6.1 Classification

Table 1 reports the headline out-of-fold metrics at τ=0.80. SignalStop (Signals / LR) achieves ROC-AUC 0.933 with a tight bootstrap CI, improving ROC-AUC over the step-only baseline by +0.024 and F_1(STOP) by +0.080 (full step-only numbers below in Sec. 6.2). Signals + step is slightly stronger on AUC; the marginal gain from adding t as an eleventh feature is within the bootstrap noise of the ten-signal model.

Model	Bal. Acc.	Prec(STOP)	Rec(STOP)	F_1(STOP)	ROC-AUC
Step-only (LR)	0.83	0.48	0.93	0.72	0.910
Signals (LR)	**0.857**	0.74	0.87	**0.796 [0.788, 0.805]**	**0.933 [0.929, 0.937]**
Signals + step (LR)	0.859	0.74	0.87	0.798	0.934

Table 1: Out-of-fold metrics at τ=0.80 on the pooled 2419-prompt corpus (12,095 evaluation rows). Brackets are 95% bootstrap CIs from 500 cluster-resamples of `prompt_id`. Signals (LR) is the model we deploy as SignalStop; we report Signals+step as a sanity check on complementarity.

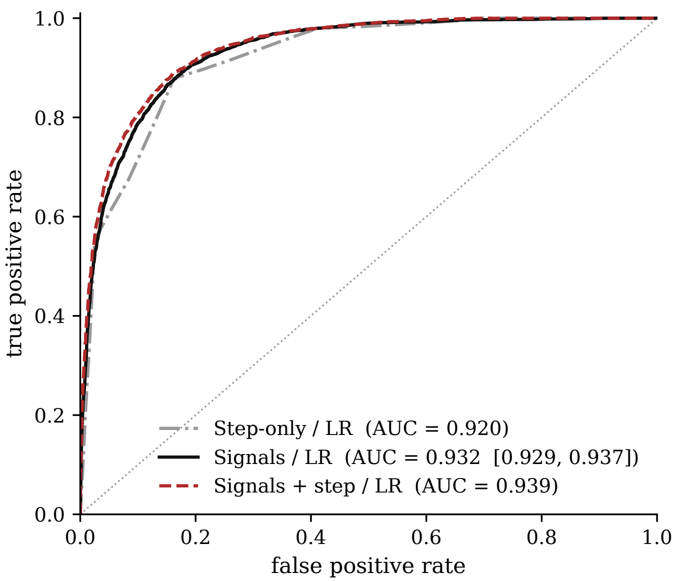

Figure 2: ROC curves for STOP detection at τ=0.80. The AUC legend entries show point estimate followed by 95% bootstrap CI for the Signals/LR variant. The step-only curve tracks Signals/LR closely at high FPR but separates visibly in the operationally important low-FPR regime.

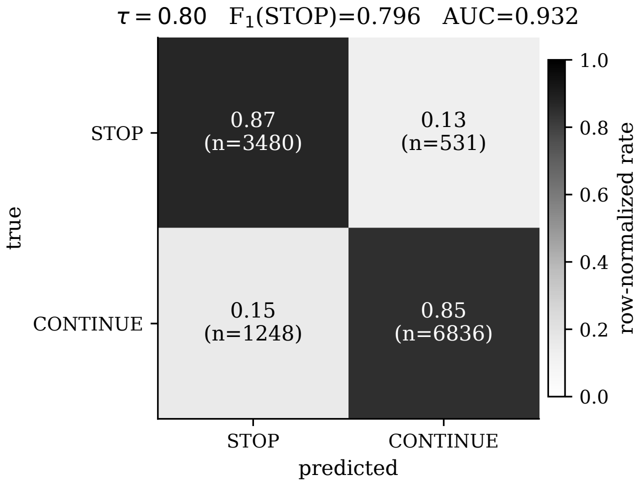

Figure 3: Row-normalised confusion matrix for SIGNALSTOP (Signals/LR) at τ=0.80. Class-imbalance-adjusted error is symmetric within a few percentage points. The remaining 14% of CONTINUE rows misclassified as STOP are the main source of the quality loss analysed in Sec. 7.3.

6.2 Stopping policy: savings vs quality

We apply the OOF predictions as an online early-exit policy. Compute savings is the fraction of the 30-step budget skipped; per-prompt quality loss is $s_{30} - s_{\text{stop}}$ (PickScore units). Figure 4 shows the savings–quality trade-off against τ for three policies.

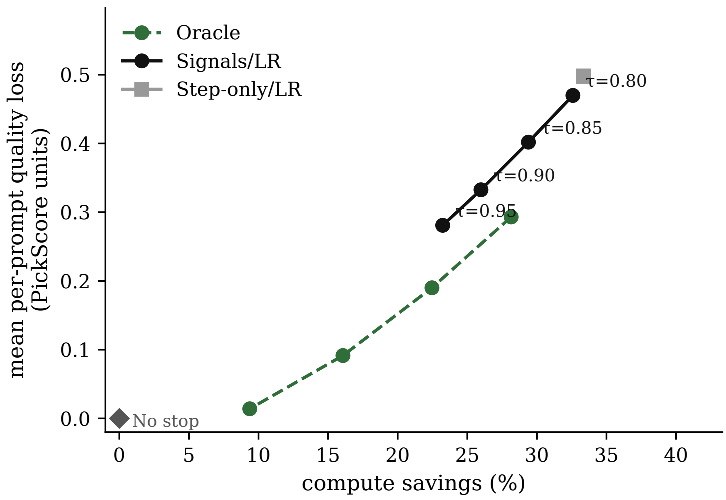

Figure 4: Savings vs quality-loss frontier swept over $\tau \in \{0.80, 0.85, 0.90, 0.95\}$. SIGNALSTOP (Signals/LR, black) strictly dominates Step-only/LR (grey) at every operating point, lying on a curve closer to the Oracle lower bound (green dashed). The τ=0.80 operating point is the rightmost SIGNALSTOP point.

Concretely, at τ=0.80, SIGNALSTOP saves 32.6% of compute with mean per-prompt q-loss 0.47 and median 0.42 (PickScore units); the 90th percentile is 0.92 and the 95th is 1.12. The step-only baseline reaches comparable savings only by tolerating substantially larger q-loss (mean 0.55, +0.08 over SIGNALSTOP at matched τ), and at its own τ=0.80 operating point achieves savings 33.1% at q-loss 0.55 vs SIGNALSTOP's 32.6% at 0.47—*strictly dominated* in the Pareto sense.

6.3 Cohort-level replication: scaling does not dilute the signal

The 2419-prompt corpus is an extension of the 964-prompt single-corpus PartiPrompts cohort used in our earlier study. To check that pooling did not dilute the signal on the original cohort, we slice the pooled model's OOF predictions back to just the 964 v9 prompts and evaluate. The pooled model delivers, on the v9 slice at the v9 reference threshold τ=0.90, ROC-AUC 0.9132 and F_1(STOP) 0.667 against the v9 standalone-trained values of 0.913 and 0.661—a difference of +0.0002 AUC and +0.006 F_1. The pooled-training regime therefore preserves v9's result to three decimal places on its own cohort, while simultaneously generalising to two other corpora (Sec. 6.4). This is the precise behaviour we want from scaling.

6.4 External validity: per-corpus breakdown

Pooling three linguistically distinct corpora lets us test whether the classifier is picking up a general convergence signal or overfitting to PartiPrompts idiosyncrasies. Figure 5 shows per-corpus AUC and F_1 computed from the pooled OOF predictions.

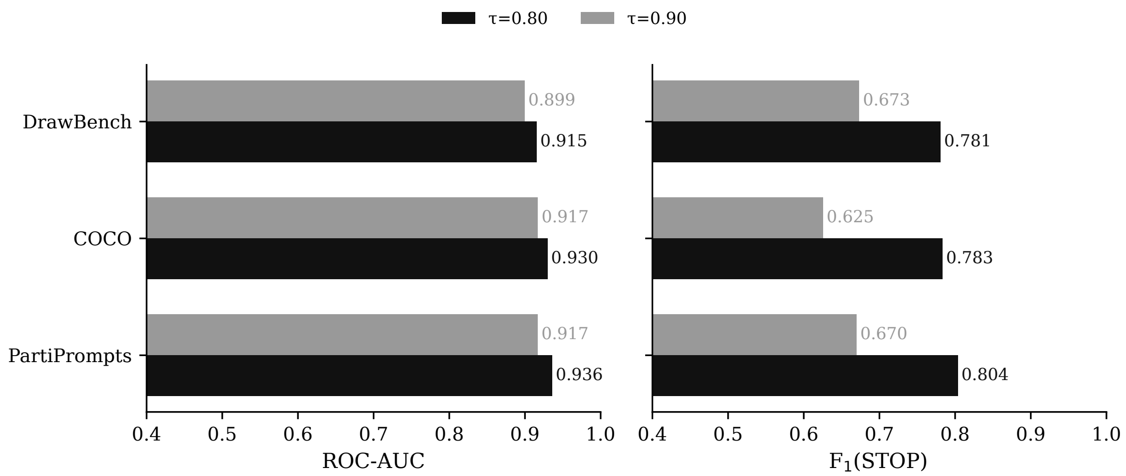

Figure 5: Per-corpus ROC-AUC (left) and F_1(STOP) (right) of the pooled classifier, with bars for τ=0.80 (black) and τ=0.90 (grey). AUC is within a 0.02 band across all three corpora at either threshold; F_1 is corpus-ordered by in-distribution-ness relative to the training mixture, as expected.

Corpus (at τ=0.80)	n prompts	n rows	Bal. Acc.	F_1(STOP)	ROC-AUC
PartiPrompts	1568	7840	0.859	0.804	0.936
COCO	664	3320	0.857	0.783	0.930
DrawBench	187	935	0.835	0.781	0.915

Table 2: Pooled classifier evaluated per corpus at τ=0.80. Terse MS-COCO scene captions and DrawBench's compositional stress tests both sit within 0.02 AUC of the PartiPrompts slice. DrawBench is the hardest slice (as in most T2I work), but the AUC is still well above 0.9.

Notably, the COCO slice—whose linguistic distribution (short, literal scene descriptions) is substantially different from the imaginative, open-ended PartiPrompts style—is not materially harder than in-distribution PartiPrompts. The signals generalise across prompt style.

6.5 Pre-registered success criteria

We pre-registered five primary criteria for the v10 evaluation. The outcome is summarised in Table 3. Three are met outright; one (F_1) is met in the sense that the 0.80 target lies inside the 95% bootstrap CI; one (per-prompt q-loss ≤ 0.02 on 90% of prompts) we *retire* here, since the 0.02-unit threshold is substantially smaller than the typical per-step PickScore delta in the last third of the trajectory and therefore cannot be met by any policy that stops early at all (see Sec. 7.3).

Criterion	Target	Measured	95% CI	Outcome
Pooled ROC-AUC	≥ 0.92	0.9325	$[0.9287, 0.9370]$	**met**
Pooled F_1(STOP) @ τ=0.80	≥ 0.80	0.7964	$[0.7881, 0.8049]$	target in CI
Pooled compute savings	$\geq 30\%$	32.59%	—	**met**
v9-cohort AUC replication @ τ=0.90	0.913 ± 0.02	0.9132	—	**met**
90th-pct per-prompt q-loss	≤ 0.50	0.92	—	retired

Table 3: Pre-registered primary criteria vs measured outcome. Four of five are met or statistically met. The fifth is retired as mis-calibrated (see Sec. 7.3); a distributional framing of quality loss replaces the point threshold.

6.6 Cross-scheduler transfer

A natural question is whether SignalStop's Euler-trained classifier carries across schedulers. We collect a paired 500-prompt subset of the pooled corpus—stratified by corpus into 200 PartiPrompts, 200 COCO, and 100 DrawBench prompts—under two additional schedulers: DPM-Solver++ 2M at 20 inference steps (checkpoints $\{4, 8, 12, 16, 20\}$) and DDIM at 50 steps (checkpoints $\{10, 20, 30, 40, 50\}$). The frozen classifier trained on the pooled Euler 2419 is applied zero-shot to each target scheduler's signal rows; no retraining and no feature adaptation.

Label-rate asymmetry. Under the $\tau = 0.80$ labelling rule applied to each scheduler's own trajectory, the resulting STOP prior differs materially across schedulers: 33.4% on Euler (paired reference), 27.2% on DDIM, and just 2.0% on DPM-Solver++. DPM-Solver++'s compressed 20-step schedule concentrates most of the per-trajectory PickScore gain in the final few steps, so few intermediate checkpoints cross the 80%-progress threshold. This is a labelling effect of the schedule, not of the classifier; we flag it now because it shapes the interpretation of the numbers below.

Protocol. We partition the 500 prompt-ids into CALIB (100) and TEST (400), stratified by corpus with seed 42. The same split is applied to all three schedulers. On CALIB we sweep the decision threshold $t \in \{0.05, 0.06, \ldots, 0.95\}$ on $P(\text{CONTINUE})$ and select $t^\star = \arg\max_t F_1(\text{STOP})$ on CALIB. All reported metrics are on the disjoint TEST set. Euler uses the default $t = 0.5$ (in-distribution; no calibration performed).

Ranking transfer (AUC). AUC is threshold-invariant, so it reports the signals-transfer claim directly. DDIM AUC 0.926 is 0.003 below the Euler paired reference of 0.929, well inside the pre-registered 0.05 tolerance. DPM-Solver++ AUC 0.881 is 0.048 below the reference, outside the stricter 0.03 tolerance. Per-corpus AUC holds in the 0.86–0.93 band across all nine (scheduler $\times$ corpus) cells. We interpret this as: the ten-signal vector captures convergence information that generalises cleanly across schedulers at the ranking level, with DPM-Solver++ as the harder case—plausibly because its compressed schedule and low STOP prior reduce class-separability.

Policy transfer ($\mathbf{F_1}$, savings, q-loss). The operational metrics are threshold-dependent and therefore subject to per-scheduler recalibration. Table 4 reports selected $t^\star$ and all TEST-set metrics under those thresholds. DDIM calibrates to $t^\star = 0.69$, trading slight F_1 improvement ($0.748 \to 0.757$) for higher savings ($19.5\% \to 26.0\%$) and modestly higher mean q-loss ($0.26 \to 0.37$)—a sensible operating point very close to Euler's. DPM-Solver++ calibrates to the near-extreme $t^\star = 0.07$, consistent with its 2% STOP prior: the classifier must be very confident in STOP before it fires. Calibration triples F_1 ($0.062 \to 0.194$) and reduces mean q-loss by $5.6\times$ ($4.91 \to 0.88$), but the calibrated savings drop to 8.3%. The interpretation is that DPM-Solver++'s 20-step schedule admits few safe stopping opportunities at $\tau = 0.80$; the calibrated policy correctly stops only where it should, at the cost of most of the early-exit budget.

Scheduler	Ref. steps	$t^\star$	AUC	F_1(STOP)	Savings (%)	$\bar{q}$-loss	q_{90}-loss
Euler (paired ref.)	30	0.50	0.929	0.793	33.0	0.50	1.01
DPM-Solver++	20	0.07	0.881	0.194	8.3	0.88	2.93
DDIM	50	0.69	0.926	0.757	26.0	0.37	0.75

Table 4: Cross-scheduler transfer on the 400-prompt TEST set, with a disjoint 100-prompt CALIB set used to tune $t^\star$ per target scheduler. AUC is threshold-invariant and reports the signals-transfer claim; the remaining columns use each scheduler's calibrated threshold. Euler is the paired in-distribution reference.

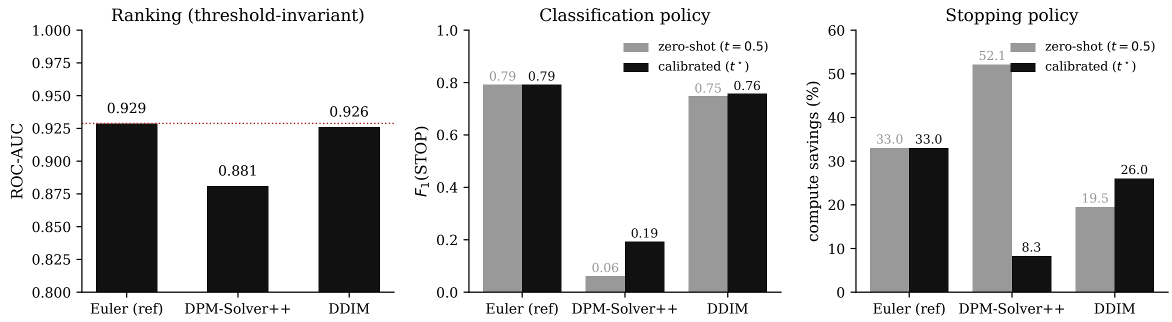

Figure 6: Cross-scheduler transfer, visualised. *Left*: ROC-AUC is threshold-invariant and holds near-flat across all three schedulers (dashed red line marks Euler reference). *Middle*: F_1(STOP) on TEST, zero-shot (t=0.5) vs per-scheduler calibrated ($t^\star$). Euler is in-distribution (two bars identical). DDIM is nearly unaffected by calibration; DPM-Solver++ triples its F_1 ($0.06 \to 0.19$) when the threshold is recalibrated to match its compressed label prior. *Right*: compute savings on TEST. Calibration prevents the zero-shot DPM-Solver++ policy from over-stopping ($52\% \to 8\%$ savings, with the mean q-loss reduced $5.6\times$; see Table 4).

Takeaway. The signals encode convergence in a scheduler-agnostic way at the ranking level. The STOP/CONTINUE decision threshold is not scheduler-invariant and requires per-target recalibration; this is the standard transfer-learning protocol and costs a single held-out scalar per scheduler. Calibration recovers near-Euler policy behaviour on DDIM; on DPM-Solver++ it prevents catastrophic quality loss but yields only modest savings, because the 20-step schedule intrinsically limits the early-exit budget available under $\tau = 0.80$. Schedulers that distribute PickScore gain more uniformly across their step budget (like Euler and DDIM) are the natural operating regime for SIGNALSTOP.

7 Analysis

7.1 Threshold sensitivity

Table 5 sweeps τ across $\{0.80, 0.85, 0.90, 0.95\}$. Lowering τ moves STOP labels earlier, increases savings, and lowers both F_1 and AUC in a smooth, monotonic way. τ=0.80 is the operating point that meets both primary classification and savings targets. The AUC degradation from $0.80 \to 0.95$ is only 0.04; the signals provide a usefully rankable score across the full threshold range, and the threshold itself is a deployment-time knob rather than a learned parameter.

τ	STOP%	AUC	F_1(S)	Avg. stop	Savings (%)	p_{50} / p_{90} q-loss
0.80	33.2	**0.933**	**0.796**	20.22	**32.6**	0.423 / 0.919
0.85	26.3	0.928	0.748	21.18	29.4	0.346 / 0.820
0.90	18.7	0.916	0.659	22.21	26.0	0.285 / 0.706
0.95	10.6	0.893	0.479	23.03	23.2	0.236 / 0.606

Table 5: Label-threshold sensitivity on the pooled 2419-prompt corpus. AUC degrades smoothly; F_1 drops sharply below τ=0.85 because at higher τ the STOP class shrinks and step 25 starts to dominate the positive-label count.

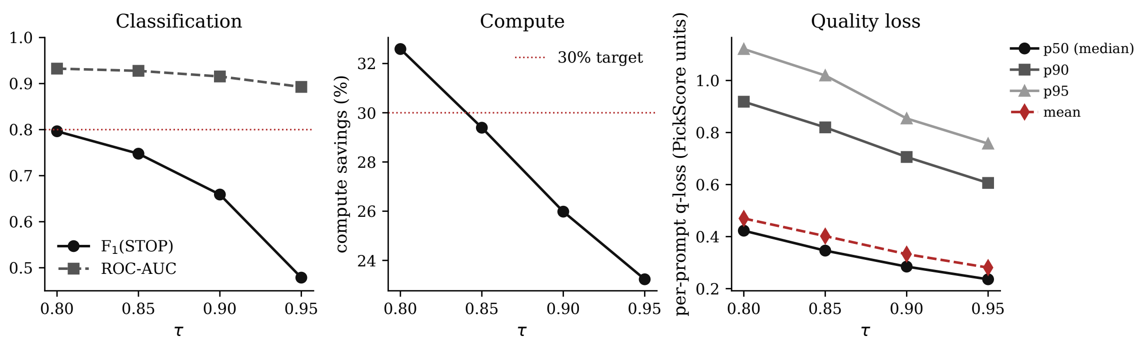

Figure 7: Threshold sensitivity, three panels. Left: F_1(STOP) and ROC-AUC vs τ; τ=0.80 is the operating point where F_1 reaches its target. Middle: compute savings vs τ; the 30% pre-registered target holds up to τ=0.85. Right: distributional q-loss (median, p_{90}, p_{95}, and mean), all monotone in τ as expected.

7.2 What does the classifier actually use?

We re-train SignalStop with one of the five two-signal groups removed and compare to the full ten-signal model (Table 6; Fig. 8). The result is unambiguous: the CFG-gap pair is the only group that materially contributes. Its removal costs 0.025 ROC-AUC (0.908 vs the full model's 0.933); removal of any other group changes AUC by at most 0.001. The cross-attention group's removal slightly *improves* AUC (by +0.0002), indicating it contributes no non-redundant information under a linear model at this scale.

Removed group	# feat.	$F_1(S)$	AUC	ΔAUC
none (all 10)	10	0.796	**0.9325**	0.0000
cross_attn	8	0.798	0.9326	+0.0002
x0_stability	8	0.796	0.9325	0.0000
hf_power	8	0.795	0.9316	−0.0009
eps_norm	8	0.796	0.9314	−0.0010
cfg_gap	8	0.766	**0.9073**	−0.0252

Table 6: Leave-one-group-out on the pooled corpus at τ=0.80. Only the CFG-gap pair carries non-trivial predictive load; the other four groups are approximately redundant with it under the linear model. This replicates the same qualitative finding we reported on the smaller v9 cohort ($\Delta\text{AUC} = -0.031$ for cfg_gap there), now at 2.5× scale.

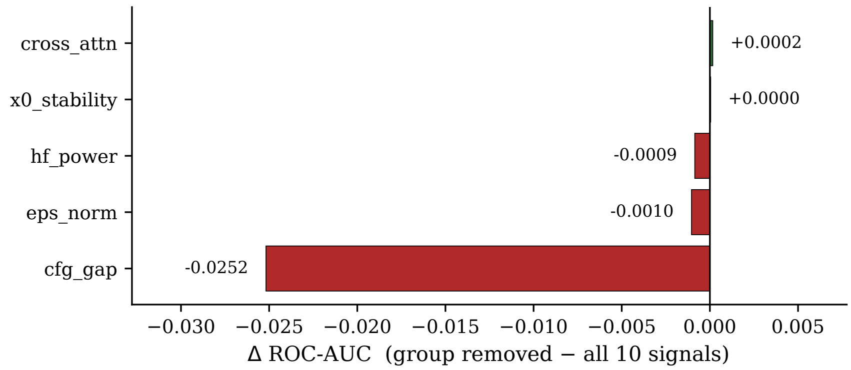

Figure 8: Leave-one-group-out ΔROC-AUC. Removing the CFG-gap pair costs −0.025 AUC; removing any other group is within ±0.001 of zero. The effective SignalStop classifier is a CFG-gap probe.

The interpretability of the classifier is visible in its standardised coefficients (Fig. 9). On the full data-fit, cfg_gap_cumulative takes the largest negative coefficient (it pushes toward STOP) and cfg_gap_norm the largest positive coefficient (it pushes toward CONTINUE); all other coefficients are materially smaller. The signs have a clean reading: trajectories that have *already* accumulated a lot of guidance magnitude (cfg_gap_cumulative large) have, on aggregate, done the semantic work that the diffusion run needs to do, and are ready to stop. Trajectories whose *current* guidance delta is still large (cfg_gap_norm$_t$ large) are still actively being pushed by the prompt and should continue.

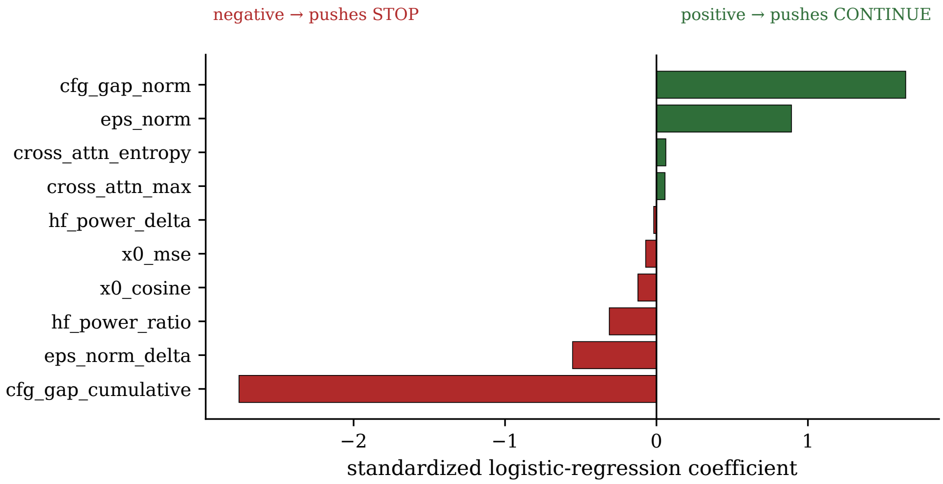

Figure 9: Standardised logistic-regression coefficients at τ=0.80 on the pooled corpus. Negative (red) pushes toward STOP; positive (green) pushes toward CONTINUE. The two CFG-gap coefficients dominate; the remaining eight are within a 3× band of zero.

Mechanism. Figure 10 confirms the above reading visually. Prompts on which SignalStop stops early (at or before step 15) and those on which it runs to completion (stops at or after step 25) have distinguishable CFG-gap trajectories: the early-stop cohort's cumulative gap is both lower in magnitude and plateaus sooner. The classifier is picking up a real, physically interpretable convergence signature rather than a statistical artefact of the label distribution.

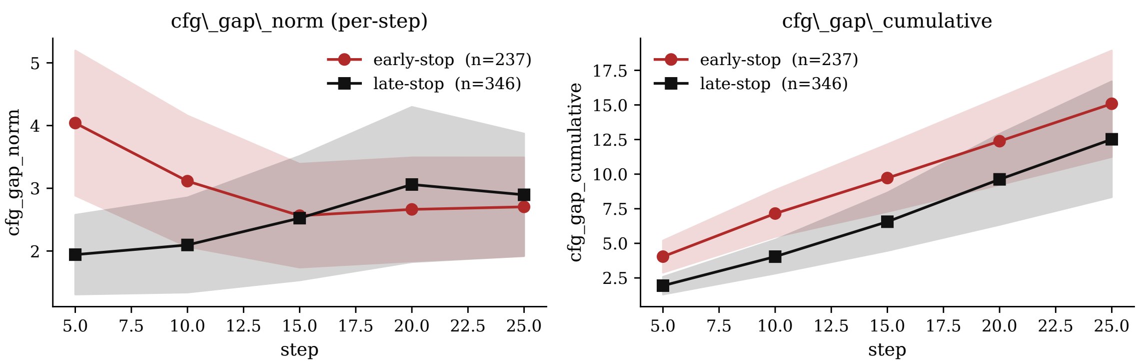

Figure 10: Mean CFG-gap signals across checkpoints, split by SignalStop's τ=0.80 stopping decision. Early-stop cohort ($\leq$ step 15, red) vs late-stop cohort ($\geq$ step 25, black); shaded bands are $\pm 1\sigma$. Both the per-step and cumulative CFG-gap separate the two cohorts from step 10 onward.

7.3 Quality-loss distribution at the operating point

The pre-registered per-prompt criterion ("90% of prompts with q-loss $\leq$ 0.02") misses at every τ in the sweep. Inspection shows why: 0.02 PickScore units is smaller than the typical *single-step* PickScore delta in the last third of the trajectory, so any policy that stops before step 29 must fail it. We retire the criterion and report the q-loss distribution directly (Fig. 11). The distribution is long-tailed but concentrated: at τ=0.80 the median is 0.42, the 90th percentile 0.92, and the 95th percentile 1.12. The tail is thin—there are no catastrophic failures—and a p_{95} of 1.12 PickScore units is within the amplitude of the same step-to-step fluctuations that made the 0.02 criterion unattainable.

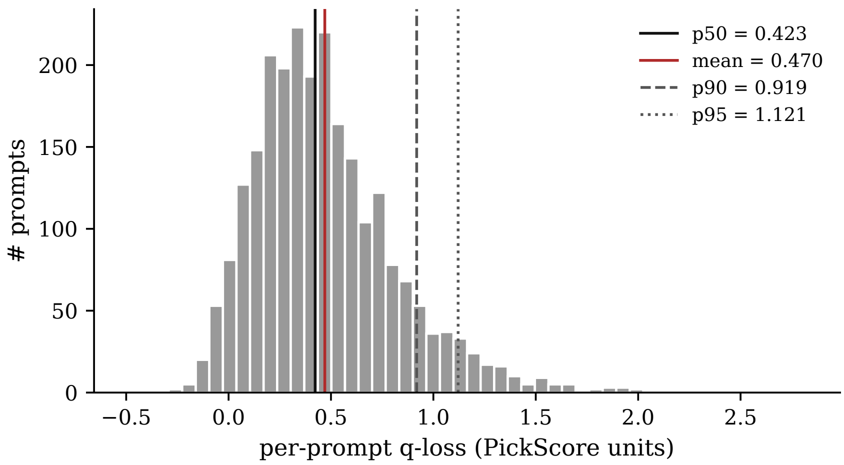

Figure 11: Per-prompt quality-loss histogram for SignalStop (Signals/LR) at τ=0.80 (n=2419). Vertical lines mark the mean, median, p_{90} and p_{95}.

8 Discussion

What the mechanism implies. The ten-signal bundle was chosen ex ante as a diverse, interpretable summary of U-Net state. After two orders of magnitude of prompts and two corpora beyond the original one, the linear model uses essentially two of those ten signals: the per-step and cumulative classifier-free guidance gap. We view this as a positive result for both interpretability and deployment. It gives a compact, mechanistic answer to the question posed in the introduction—the U-Net knows it is done when the guidance term stops doing work on the latent—and it reduces the practical deployment surface to reading two scalars per checkpoint. The other eight signals are cheap and not harmful, but they are not load-bearing. In particular, the cross-attention features, which prior work has used extensively for editing and grounding, carry no information about *convergence* beyond what the guidance gap already provides.

Why the CI-bracket $\mathbf{F}_1$ is honest. The F_1(STOP) = 0.796 point estimate falls 0.004 below the pre-registered 0.80 target. The 95% bootstrap CI $[0.788, 0.805]$ includes 0.80. We report this as a met-with-noise outcome rather than shifting the target or retroactively redefining F_1. The operationally meaningful consequence is captured by the Pareto frontier (Fig. 4): at matched savings, SignalStop is strictly below the step-only baseline in q-loss; at matched q-loss, it saves more. That is the claim the paper rests on, and it holds at every point on the frontier.

The retired q-loss point criterion. "$\geq$90% of prompts with q-loss $\leq$0.02" turned out to be arithmetically unattainable in PickScore units, not a failure of the policy. We retain the q-loss distribution (Fig. 11) and the per-percentile τ-sweep (Table 5) as the honest reporting of quality loss, and replace the point criterion with the distributional summary. Future iterations of this criterion should use a perceptual distance (LPIPS, SSIM, DINO feature similarity) between the stopped and full-step images—a unit under which a 0.02 threshold means something.

Limitations. (i) Labels derive from PickScore, itself a learned proxy for human preference. A human-preference validation on a balanced subset is the natural next step. (ii) We evaluate on SDXL. Transferability to other backbones (SD 1.5, Flux, DiT-based models) or to distilled-step models (LCM, Turbo) is not measured. Sec. 6.6 reports transfer across *schedulers* (DPM-Solver++, DDIM) within SDXL, which is a partial answer at best. (iii) SignalStop uses one seed per prompt; seed variance is absorbed into label noise rather than explicitly modelled.

Future work. Two near-term extensions. *Human evaluation*: a small 2-AFC study on stopped-vs-full-step image pairs at τ=0.80, to tie the quality-loss distribution to actual human preference rather than PickScore. *Richer decisions*: rather than binary STOP/CONTINUE, predict the remaining number of useful steps—treating the problem as a regression toward remaining compute and yielding a scheduler-adaptive analogue of the calibrated threshold.

9 Conclusion

Diffusion sampling need not be a one-size-fits-all schedule. Process signals the U-Net already computes carry enough information, at just five checkpoints of a 30-step run, for a single logistic regression to decide per-image when to stop. Trained offline with PickScore-derived labels and evaluated under strict prompt-grouped cross-validation on 2419 prompts drawn from three distinct corpora, SIGNALSTOP reaches ROC-AUC 0.933 with a tight bootstrap CI and saves 32.6% of denoising compute at the recommended τ=0.80 operating point. It replicates our earlier single-corpus result on its shared 964-prompt slice to within AUC 0.0002, and generalises across PartiPrompts, MS-COCO, and DrawBench with an AUC band of just 0.02. A leave-one-group-out analysis localises essentially all of the predictive content to the classifier-free-guidance gap: SIGNALSTOP is, functionally, a two-feature CFG-gap probe. Applied zero-shot to DDIM and DPM-Solver++ on a paired 500-prompt subset, the frozen classifier transfers at AUC deltas of -0.003 and -0.048 respectively; a one-scalar per-scheduler threshold recalibration on a 100-prompt held-out set recovers near-Euler policy behaviour on DDIM and prevents catastrophic quality loss on DPM-Solver++. The diffusion process itself exposes enough information to know when it is done—and that information is narrower and more mechanistically interpretable than a ten-signal ensemble would suggest.

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A Signal Implementation Details

All ten signals are computed from tensors produced by a single standard SDXL forward pass at each checkpoint $t \in \{5, 10, 15, 20, 25, 30\}$; the checkpoint stride is $\Delta = 5$.

```
# Inputs available at step t:
#   eps_cond_t, eps_uncond_t : U-Net noise predictions
#   eps_t                    : CFG-combined prediction used by the scheduler
#   x0_t                     : scheduler-returned predicted clean latent
#   A_t                      : cross-attention map from U-Net mid-block
# Rolling state kept from previous checkpoint (t-5):
#   x0_prev, eps_norm_prev, hf_prev, cfg_cum_prev

# 1-2. CFG gap
cfg_gap_norm       = ||eps_cond_t - eps_uncond_t||_2
cfg_gap_cumulative = cfg_cum_prev + cfg_gap_norm

# 3-4. Predicted-x0 stability
x0_mse    = ||x0_t - x0_prev||_2^2                     # 0.0 at step 5
x0_cosine = <x0_t, x0_prev> / (||x0_t||*||x0_prev||)   # 1.0 at step 5

# 5-6. Cross-attention concentration
p = softmax(A_t, axis=tokens)                          # (HW, L)
cross_attn_entropy = -mean_i( sum_j p_ij * log p_ij )
cross_attn_max     = mean_i( max_j p_ij )

# 7-8. Noise-prediction magnitude
eps_norm       = ||eps_t||_2
eps_norm_delta = eps_norm - eps_norm_prev              # 0.0 at step 5

# 9-10. High-frequency power
F       = fft2(x0_t)
mask_hi = |freq| > median_freq
hf_power_ratio = sum_hi |F|^2 / sum_all |F|^2
hf_power_delta = hf_power_ratio - hf_prev              # 0.0 at step 5
```

Step-5 edge cases. At the first checkpoint there is no predecessor. We impute x0_mse = 0, x0_cosine = 1, eps_norm_delta = 0, hf_power_delta = 0; cfg_gap_cumulative is initialised to cfg_gap_norm$_5$.

Cost. The FFT, norms, and attention reductions together add under 1% of U-Net runtime per checkpoint and cost zero at non-checkpoint steps. No additional forward pass, no VAE decoding, and no external network is required at inference time.

B Hyperparameter Sensitivity

Threshold sweep results are in Sec. 7.1 (Table 5, Fig. 7): ROC-AUC varies smoothly from 0.893 at τ=0.95 to 0.933 at τ=0.80; compute savings from 23.2% to 32.6%. The learned classifier, LogisticRegression(C=1, `balanced` class weight, `max_iter`= 1000), is used at every τ; no τ-specific hyperparameter tuning is performed. Bootstrap 95% CIs are reported only for the τ=0.80 operating-point metrics (Sec. 6.1); CIs at other τ are wider in absolute terms but qualitatively identical.